Validity-Aware Active Semantic Exploration

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Abstract

Active Semantic Perception lets a mobile robot complete its partial scene graph with language model predictions and use them to select viewpoints. This project reproduces that pipeline with a substitute language backend and tests three extensions: whole-graph validation of model completions, calibrated ensemble support filtering, and a risk-sensitive viewpoint score. The final evidence supports a smaller conclusion than the proposal. On three indoor scenes at a 120 m path budget, keeping only predictions supported by all four ensemble members improves static graph quality, with object F1 gains of 8.3 to 9.3 percent and normalized graph edit distance reductions of 7.3 to 8.0 percent, while removing about 27 percent of predicted nodes at negligible FULL-GRAPH true-positive cost; this does not establish predicted-only true-positive retention. Calibration matches raw thresholding at full unanimity and is harmful at intermediate thresholds on one scene. The risk term is either inactive or counterproductive. Decision replay shows a narrow descriptive filtering effect without establishing any live navigation benefit. The study is presented as a careful replication with negative results, with all artifacts, ledgers, and corrections preserved.

Keywords: active semantic perception, scene graphs, ensemble filtering, uncertainty calibration, negative results, robot exploration.

I. Introduction

Robots that explore indoor spaces build maps as they move. Active Semantic Perception (ASP) adds a semantic layer: it maintains a graph of rooms and objects, asks a language model to predict missing structure, and scores candidate viewpoints by how much they would reduce uncertainty [1]. The language model can invent objects that are not there, and those inventions can steer the robot badly. This project asks whether checking and filtering those predictions makes them more useful for exploration.

We reproduce the ASP pipeline, add a fail-closed graph validator, ensemble-support filtering, leave-one-scene-out calibration, and a risk-sensitive viewpoint score, then measure each addition against the unmodified baseline. The experiments are frozen and the final interpretation is deliberately narrower than the proposal. Full ensemble unanimity is a useful static graph diagnostic on the tested backend; calibration and the risk term are not positive contributions; decision replay is informative but cannot establish live navigation improvement. The remainder of this paper describes the reproduction scope, the method, the experiments, and the limits of what the evidence supports.

II. Background and Reproduction Scope

The work builds on the public ASP implementation at the pinned commit f1ea141b, with documented working-tree deviations preserved alongside it [1]. The pinned Gemini backend was deprecated during the project and its replacement required billing that was not enabled, so the study reimplements the ASP method with a substitute backend (DeepSeek through a metered proxy) rather than replicating the published numbers. All policy comparisons run on the same backend and stay internally valid, but no claim about the original backend follows.

Ground-truth references are self-generated by running the language-model-free frontier baseline to exhaustive coverage at zero model cost. Four indoor scenes were staged using HM3D [2]; the final matrix covers three scenes (00069, 00573, 00853) at seed 42 under a 120 m path budget. Scoring uses two evaluators that are not interchangeable: an author-faithful tracked-graph evaluator with greedy same-name matching and NetworkX graph edit distance [3], and a canonical static evaluator with globally distance-sorted matching and a node-plus-edge edit proxy. Reported graph edit distances from the author-faithful evaluator are best-found upper bounds under a 60 s solver timeout, not certified exact values.

III. Method

A. Canonical Adapter and Structural Validator

Author scene graphs serialize geometry as strings and edges as pairs, so a canonical adapter converts them to one schema and derives object-to-room parents. The validator then checks each model completion: it rejects completions that are unparseable or that drop observed nodes, and it removes predicted nodes that fail geometric checks (overlap with observed

structures, missing parent rooms, crossings of impassable structures without doors). Reference labels are attached only post hoc for diagnostics and never feed validation decisions. An ensemble in which every member is rejected falls back to the observed map with deterministic frontier faces.

B. Ensemble Support Policy

Each scene track is completed by its own four-member ensemble. Raw support for a predicted node is the fraction of the intended four members containing a matching hypothesis, never the fraction of surviving members, so parsing failures cannot inflate support. The threshold grid is limited to the reachable values {0, 0.5, 0.75, 1.0}. Full unanimity (tau = 1.0) keeps only predictions that all four members support.

C. Calibration and Risk Score

A leave-one-scene-out isotonic calibrator maps raw support to calibrated probability, pooled across room and object hypotheses. Every replay record carries an explicit calibration status, and calibrated rows are read only when the status is LOSO_CALIBRATED. The risk-sensitive score re-ranks the shortlist by calibrated uncertainty with a weight beta; calibration can reach decisions only through the support filter, so the two mechanisms are never double-counted on the same nodes.

D. Evaluation Tracks

Three analyses read the frozen artifacts. Static replay filters the shared cached completions under each policy and integrates quality over path distance up to the budget. Decision replay replays viewpoint selection with the pinned uncertainty calculator over one shared candidate pool per checkpoint and reports selection displacement and reference visibility. Phase 13 diagnostics combine calibration, validator, guardrail, and availability checks and fail closed on missing inputs.

IV. Experimental Setup

Short-run baselines on scene 00069 established the pipeline: a low-effort 25 m run reached tracked-graph area-under-curve precision 0.671, recall 0.241, F1 0.352, and normalized graph edit distance 1.075, while a high-effort comparison at matched budget performed worse (F1 0.302), a configuration result rather than a clean causal estimate of reasoning effort. The final matrix ran all three scenes past 120 m (125.3 m, 121.9 m, and 126.1 m) and all headline numbers are path-normalized areas capped at exactly 120 m with the canonical evaluator. A separate Gemini campaign exercised the metered request infrastructure across nine scene and seed combinations but reached only 19.5 m in its final usable block before reservation accounting stopped it; it is reported separately and merged with nothing.

The local repository is an implementation and selected-evidence package. Full simulator state, datasets, and raw run directories remain in the remote workspace; the mirror holds selected results plus a small cached example. Static replay resolves calibration folds by exact workspace reference paths, so results inspected from a relocated clone must be read with their recorded calibration status.

V. Results

A. Static Unanimity Filtering

Full ensemble unanimity improves static graph quality on all three scenes while recall stays flat, as Table I shows. Preregistered guardrails pass everywhere: absolute F1 loss is at most 0.05 and normalized graph edit distance increases by no more than 10 percent relative to the baseline. The observed GED decrease is a result, not a gate.

TABLE I. Static Unanimity Results (120 m-Capped AUC, Canonical Evaluator)

Scene	Official F1	Unanimity F1	Official GED	Unanimity GED
00069	0.4789	0.5218	0.8969	0.8254
00573	0.3980	0.4349	0.9576	0.8877
00853	0.4578	0.4957	0.8561	0.7930

B. Mechanism and Validator Role

The filter removes about 27 percent of predicted nodes per stage while retaining about 99.7 percent of full-graph evaluator true positives. That retention figure is dominated by the unchanged observed map; predicted-only true positives fall (objects 61

to 45, rooms 313 to 138 in the validator diagnostics). The near-constant precision lift follows mechanically: true positives stay nearly flat while the prediction denominator shrinks. The structural validator alone beats the baseline slightly, but on top of unanimity it adds nothing measurable, so unanimity subsumes what the validator catches rather than the validator doing nothing. Because untagged predicted rooms are invisible to both policies, this is an object-level filtering result. Consistency across scenes reads as a stable ensemble-disagreement rate, a property of the tested backend rather than demonstrated generalization across environments.

C. Calibration

Calibration coincides exactly with raw thresholding at full unanimity on every scene, so the recommended policy is unaffected, but it is not neutral elsewhere. On scene 00573 its fold maps three raw support levels to one calibrated value, collapsing two working filters into no filtering, as Table II shows. The training signal explains why: only 23 positives exist in 2,829 held-out rows, with 16 room rows, so the calibrator cannot separate adjacent support levels for the object population that matters.

TABLE II. Calibration Harm at Intermediate Thresholds (Scene 00573, F1)

Policy	tau 0.5	tau 0.75
Raw support	0.4311	0.4345
Calibrated	0.3978	0.3978

D. Risk-Sensitive Score

Calibrated support probabilities sit near the floor almost everywhere, so at beta of 1 or below the combined policy is bit-identical to the baseline, while at beta 5 or 20 the only surviving signal rewards looking where nothing is predicted. The term is therefore either inactive or counterproductive, and the named fix was deliberately left unevaluated to avoid in-sample selection.

E. Decision Replay

Within the deterministic fixed-pose approximation, structural filtering moves selected viewpoints toward more novel reference objects on all three runs, with a moderate descriptive margin on two scenes and no useful signal on the largely blind third, as Table III shows. An identical-graph null matches the baseline exactly at all 94 stages with zero displacement, which rules out calculator-path variation but measures neither seed noise nor live reachability. Stages are sequential and autocorrelated, so win and loss counts are descriptive, not independent trials.

TABLE III. Novel Reference Objects, Official to Filter-Only (Descriptive)

Scene	Official	Filter-Only	Record
00069	25	50	12-4
00573	15	23	7-4
00853	40	64	13-4

F. Guardrails and Availability

Phase 13 records 752 expected completion slots, 686 available, 667 accepted, 19 rejected, and 66 missing, with the unanimity guardrail policy passing on all three scenes. The false-positive detour gate is not identifiable, robot collisions are not instrumented, and combined live usefulness is not established. These are hard limits on what the evidence can claim, not gaps in reporting.

G. Partial Gemini Block

The final Gemini block reached 19.5 m over seven completed stages and one partial stage. Its tracked graph reached precision 0.692, recall 0.336, F1 0.452, and normalized graph edit distance 1.109 at stage 6; the best static completion policy reached precision 0.769 and F1 0.284. All 55 available completions were accepted with 35 predicted nodes removed. The block settled 898 requests for $18.552664 within $65.749854 of summed reservations, which is why the shared allocation stopped. These are static prediction replays, not a live counterfactual trajectory.

VI. Limitations

One seed cannot separate a real effect from seed noise, and three fixed scenes with autocorrelated stages cannot establish generalization; no confidence intervals are reported at this sample size. The author checkout carries documented working-tree deviations, and the host supervisor record closed in a paused state with a post-budget traceback on the third scene that was analyzed and left untouched. The audit trail also records a prospective policy path in which echoed observed identifiers can duplicate during support filtering; the canonical replay uses marked graphs and a separate normalization path; it is not automatically invalidated, but no blanket unaffected claim is warranted. Recorded Gemini settled cost is $37.353274 across five ledgers, with three small probe entries unresolved and counted as unknown rather than zero.

VII. Conclusion

The project contributes a careful replication with one supported static finding and two documented negative results. Requiring full ensemble agreement removes mostly false-positive predictions at negligible FULL-GRAPH true-positive cost on the tested backend; predicted-only true-positive retention was not established. Calibration does not improve the useful endpoint and harms intermediate settings; the risk term does not help viewpoint selection. Decision replay is consistent in direction but cannot speak to live navigation. All code, selected evidence, ledgers, and corrections are preserved so that the boundary between what was shown and what was not stays checkable.

Acknowledgment

Compute ran on a shared GPU host; the metered campaign safeguards and the host supervisor kept paid usage inside preregistered budgets.

References

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